

Evaluating Tokyo’s Cap-and-Trade Emissions Trading System

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Introduction

We are analyzing the effectiveness and design of metropolitan Tokyo’s Cap-and-Trade Emissions Trading System and how it has evolved over time. This is one of the world’s first urban emissions trading programs; by studying its environmental and economic outcomes, we seek to learn how market-based policies can affect decarbonization in dense metropolitan areas.

Goals

- 1. Understand the **design** of Tokyo’s Cap and Trade System and how it regulates large commercial and industrial facilities.
- 2. **Evaluate emissions reduction** in the first two compliance periods
- 3. **Examine energy-efficiency and renewable-energy** responses
- 4. **Assess economic impacts** specifically employment levels
- 5. Identify **broader policy responses** for emissions trading systems

History

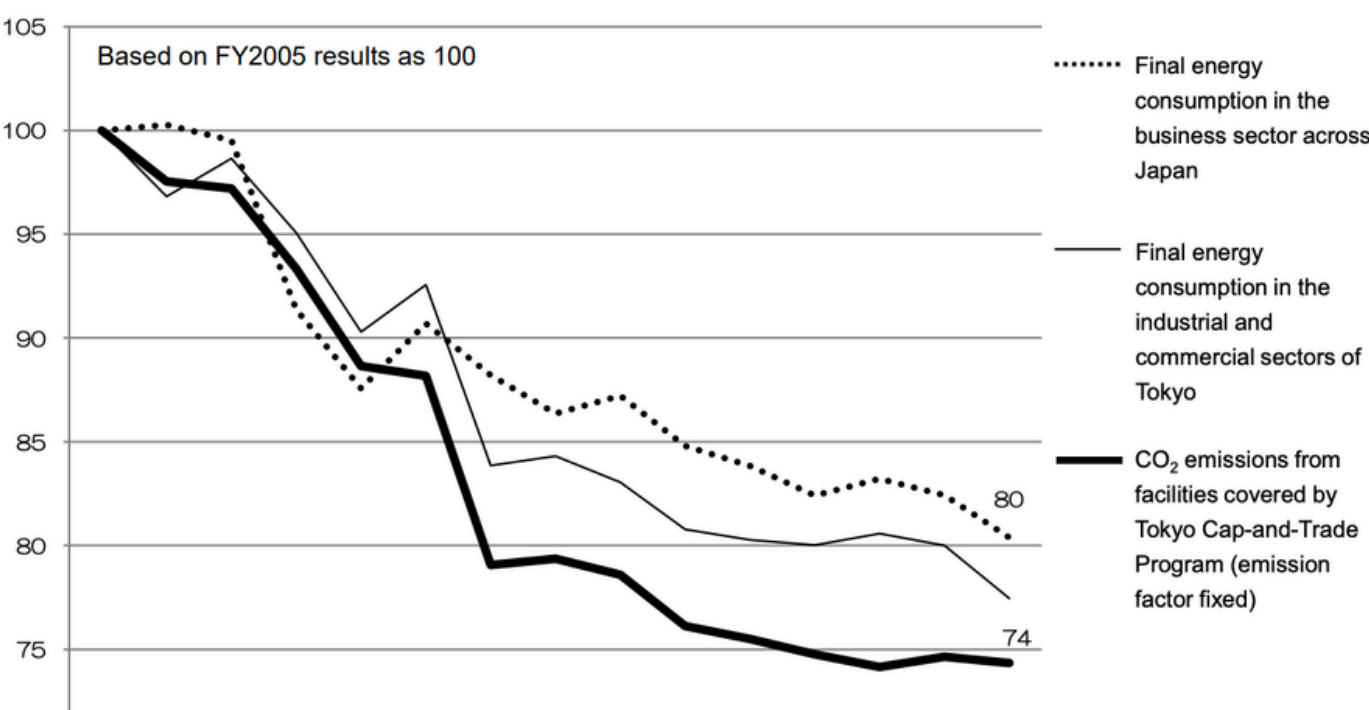
Tokyo's cap-and-trade system was brought to the Tokyo Metropolitan assembly in 2008 and launched in 2010, exploring a comprehensive plan for Tokyo to fight global warming, complete with reporting systems, verification and enforcement. The Tokyo Emissions Trading System (ETS) uniquely targets office and commercial buildings, making it the world’s first emissions trading scheme to comprehensively cover the urban service sector.^[1]

Policy Scope:

- **Target:** Required CO₂ reductions for large commercial and industrial buildings that consume >=1,500 KL of energy per year
- **Coverage:** Affects 1,300 facilities (1,000 office buildings and 300 factories)^[2]
- **Impact:** Encapsulated around 40% of CO₂ emissions in commercial area of Tokyo

Emitters could either meet targets by decreasing emissions or participating in an emissions trading program via credits in the form of:

- 1. Excess emission reductions
- 2. Small and mid-size facility credits
- 3. Renewable energy credits
- 4. Outside Tokyo credits
- 5. Saitama credits (average cost of 600 JPY or 3.96 USD per unit)^[6]



Key Findings

Phase I (2010–2014)

- Facilities achieved a **12.7% emissions reduction** relative to baseline, or a reduction of 12.27 million tons of CO₂-equivalent.^[3]
- However, econometric analysis shows that ~1/2 of the observed reduction (~6.9%) was directly attributable to the Tokyo ETS, while the other half (~6.4%) resulted from higher electricity prices following the 2011 Great East Japan Earthquake.
- Broader citywide energy consumption decreased by 21.1% from 2000-2015, with notable declines in **industry (–40.8%)** and **transport (–41.7%)**, alongside an 8.5-fold rise in **solar capacity** supporting renewables’ share exceeding 11% of electricity use.
- **Transportation** led reductions, achieving a 36.1% decline over 2000–2015 through mass transit expansion and fuel-efficiency programs in freight fleets.^[3]

Phase II (2015-2018)

- Facilities achieved a **27% emissions reduction** in emissions relative to baseline, or 21.9 million tonnes of CO₂-equivalent.
- Around 80% of facilities exceeded their mandatory reduction targets (by ~15-17%), indicating widespread compliance.

Energy-Efficiency and Renewable-Energy Responses

- Majority of reductions achieved through **direct energy-efficiency upgrades**, including heat, air-conditioning, and lighting systems.
- Key measures include high-efficiency pumps for air conditioning systems (~148,000 t-CO₂), high-efficiency lighting installations, notably LED lighting and smart controls (~158,000 t-CO₂), etc.^[5]
- Facilities are increasingly planning for long-term renewable energy integration—including corporate power purchase agreements and on-site renewable installations—in preparation for Tokyo’s 2030 and 2050 decarbonization targets.
- Respondents also highlighted expectations for government support in technological innovation and renewable supply stability, showing **policy-driven behavioral shifts** beyond compliance.^[5]

Economic Impacts

- No statistically significant negative impact on employment among regulated firms in neither Phase I nor II.^[4]
- The Tokyo ETS did not harm nor meaningfully boost business activity or competitiveness, supporting the fact that carbon regulation can **coexist** with **stable employment** levels.^[4]

Reduction measures indicated in GHG Emission Reduction Plans		
Measure for heat sources, air conditioning and lighting	Quantity	Reduction (tonnes)
Installation of high-efficiency lighting & energy-saving control	2,193	158,027
Installation of high-efficiency heat source equipment	420	148,309
LED lights included in above	1,913	138,266
Installation of high-efficiency air conditioning equipment	438	39,616
Installation of high-efficiency pumps for AC & energy-saving control	340	31,618
Installation of systems for cooling using outside air	261	23,163
Relaxing illumination conditions	213	16,503
Installation of high-efficiency fans	244	14,072
Installation of external air volume control based on CO ₂	128	14,481
Hf lights included in above	147	13,001
Other (all remaining measures)	N/A	912,833
Total	N/A	1,428,669

Policy Implications

1. ETSs generate **broader institutional benefits** by improving infrastructure, and strengthening mechanisms for monitoring, reporting and verification, which enhance transparency and accountability.^[3]
2. While external variables, such as energy price fluctuations, also contributed to reductions, the Tokyo ETS demonstrates that well-designed trading systems can **amplify** and **sustain** these effects, serving as **complementary policy instruments** reinforcing initiatives.
3. Participation in urban emissions trading systems have **no significant adverse effects on employment or industrial activity**.
4. Statistics on emissions reduction **signal** towards the program’s long term success, providing institutions with the **certainty** and **results** needed for further regulation and investments in future endeavors such as cleaner technology, and stronger infrastructure.
5. ETS can indirectly help foster greater **education** and **awareness** of climate change, better preparing stakeholders for more effective participation in future climate initiatives.^[4]

Conclusion & Looking Forward

- Tokyo’s ETS has demonstrated strong success, achieving more than **27% emissions reduction** in Phase II and **surpassing mandatory targets** across most regulated facilities.
- **Lowering the emissions cap** in future phases would build on this momentum and encourage deeper, long-term decarbonization.
- Increased investment in **clean energy infrastructure**—such as renewable procurement, on-site solar, and next-generation building efficiency technologies—will be essential for sustaining progress toward Tokyo’s 2030 and 2050 climate goals.
- Introducing **new incentive structures** (e.g., subsidies, credit bonuses, or performance rewards) could encourage faster adoption of innovative low-carbon technologies beyond what cap-and-trade mandates alone can achieve.
- Tokyo’s successful implementation provides a **transferable model** for other global cities, showing that well-designed urban ETS programs can reduce emissions effectively without harming economic performance.

References

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